

LIMIT-SKETCHABLE INFINITY CATEGORIES

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Presentability

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- ✓ C admits κ -filtered colimits.
- ✓ There is a set of κ -compact objects that generate C under κ -filtered colimits.

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Presentable	Accessible not presentable	Not accessible
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Set		
Grp		
sSet		
RMod		
Ch(RMod)		
Sheaf topos		

	Fields	
	Hilbert spaces	

		Top
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Limit sketches

A **limit sketch** (Bastiani and Ehresmann 1972) is a pair (A, C) of a small category A and a set of cones C over A .

Example. Let A be the small category generated by the square (a).

$$\begin{array}{ccc} 0 & \longrightarrow & 1 \\ \downarrow & (a) & \downarrow \\ 2 & \longrightarrow & 3 \end{array}$$

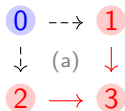
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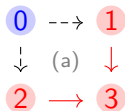
A **model** of a limit sketch is a functor $F : A \rightarrow \mathbf{Set}$ which sends cones of C to limits of **Set**. A category is **limit-sketchable** if it is equivalent to the category of models of some limit sketch.

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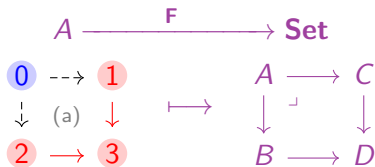
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A model F of the sketch $(A, \{c\})$ is a pullback of sets 🍑

Representation theorem

Theorem (Adamek and Rosicky 1994)

The following are equivalent:

- (i) *Presentable categories.*
- (ii) *Accessible localizations of presheaves over a small category.*
- (iii) *Limit-sketchable categories.*

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Goal

Presentable ∞ -categories $\stackrel{?}{\simeq}$ Limit-sketchable ∞ -categories

Plan

Presentable ∞ -categories

Limit ∞ -sketches

Representation theorem

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Review of higher categories

Informally, an ∞ -category (or $(\infty, 1)$ -category) has objects and:

- n -morphisms between $(n - 1)$ -morphisms $\forall n \geq 1$,
- $\circ$ and id are weak up to a higher morphism,
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In the Joyal-Lurie model of higher categories:

∞ -**category** := Quasicategory (sSet with weak composition)

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Example

- (a) Any 1-category can be viewed as an ∞ -category using the nerve functor $N : \mathbf{Cat} \rightarrow \mathbf{sSet}$.
- (b) The ∞ -category of homotopy types $\mathcal{S}$.

Limits and colimits

Let $\mathcal{C}$ be an ∞ -category, and I be a small simplicial set. Given any object $x \in \mathrm{Obj}(\mathcal{C})$, the **constant diagram** $\Delta_x : I \rightarrow \mathcal{C}$ sends all objects of I to x , and all higher morphisms to higher identities.

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Let $D : I \rightarrow \mathcal{C}$ be a diagram and $y \in \text{Obj}(\mathcal{C})$ be an object of $\mathcal{C}$. A natural transformation $\alpha : \Delta_y \Rightarrow D$ **exhibits y as a limit of D** if, for all $x \in \text{Obj}(\mathcal{C})$, α induces an equivalence

$$\text{Map}_{\mathcal{C}}(x, y) \xrightarrow{\sim} \text{Cones}(I, D, x) := \text{Map}_{\text{Fun}(I, \mathcal{C})}(\Delta_x, D).$$

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Cocones and **colimit cocones** are defined as cones and limit cones in the opposite ∞ -category.

Size of ∞ -categories

Let κ denote a regular cardinal and $\mathcal{C}$ an ∞ -category.

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- $\mathcal{C}$ is **locally small** if for all objects $x, y \in \text{Obj}(\mathcal{C})$ the mapping space $\text{Map}_{\mathcal{C}}(x, y)$ is essentially small.

Accessibility

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- An object $x \in \text{Obj}(\mathcal{C})$ is called κ -**compact** if the mapping space functor $\text{Map}_{\mathcal{C}}(x, -) : \mathcal{C} \rightarrow \mathcal{S}$ preserves κ -filtered colimits. Denote by $\mathcal{C}_{\kappa}$ the full sub- ∞ -category of κ -compact objects.

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An ∞ -category $\mathcal{C}$ is **accessible** if it is locally small and there is a regular cardinal κ such that:

- ✓ $\mathcal{C}$ admits κ -filtered colimits.
- ✓ $\mathcal{C}_{\kappa}$ is essentially small and generates $\mathcal{C}$ under κ -filtered colimits.

Presentability

Definition

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Example

- (a) $\mathcal{S}$ is presentable.
- (b) Any ∞ -topos is a presentable ∞ -category.
- (c) Any presentable 1-category is also a presentable ∞ -category.
- (d) If $\mathcal{A}$ is a small ∞ -category and $\mathcal{C}$ is a presentable ∞ -category, then $\mathrm{Fun}(\mathcal{A}, \mathcal{C})$ is presentable.
- (e) In particular, $\mathrm{PSh}(\mathcal{A}) := \mathrm{Fun}(\mathcal{A}^{\mathrm{op}}, \mathcal{S})$ is presentable.

Characterization of presentability

A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is a **localization** if it has a fully faithful right adjoint:

$$\mathcal{D} \begin{array}{c} \xleftarrow{F} \\ \perp \\ \xrightarrow{U} \end{array} \mathcal{C}$$

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If $F : \mathcal{C} \rightarrow \mathcal{D}$ is a localization, then $\mathcal{D} \simeq \mathcal{L}_W \mathcal{C}$ where

- $\mathcal{L}_W \mathcal{C}$ is the ∞ -category that **universally inverts** W .
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Let $\mathcal{C}$ be a complete ∞ -category. A functor $F : \mathcal{K} \rightarrow \mathcal{C}$ is a **model** of a limit ∞ -sketch $\mathcal{T} = (\mathcal{K}, \mathfrak{L})$ in $\mathcal{C}$ if it sends each cone in $\mathfrak{L}$ to a limit cone in $\mathcal{C}$.

$\mathrm{Mod}(\mathcal{T}, \mathcal{C}) := \infty\text{-category of models of } \mathcal{T} \text{ in } \mathcal{C}$

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We say that an ∞ -category is **limit ∞ -sketchable** (or **essentially ∞ -algebraic**) if it is equivalent to the ∞ -category of models of some limit ∞ -sketch.

Examples: Pointed objects

Let $\mathcal{A}$ be the nerve of the small category generated by the graph

$$0 \xrightarrow{a} 1 ,$$

and $\mathfrak{L}$ be the set of a cone with the empty diagram $D : \emptyset \rightarrow \mathcal{A}$, the apex $0 \in \mathcal{A}$, and the unique natural transformation $\alpha : \Delta 0 \Rightarrow D$.

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Then $\mathcal{T} = (\mathcal{A}, \mathfrak{L})$ is a limit ∞ -sketch, and a model $F : \mathcal{A} \rightarrow \mathcal{C}$ sends

$$(a : 0 \rightarrow 1) \longmapsto (F(a) : 1_{\mathcal{C}} \rightarrow F(1)),$$

where $1_{\mathcal{C}}$ is the terminal object of $\mathcal{C}$.

Therefore, $\text{Mod}(\mathcal{T}, \mathcal{C})$ is the ∞ -category of **pointed objects** in $\mathcal{C}$.

Examples: ∞ -algebraic theories

An **∞ -algebraic theory** (or **∞ -Lawvere theory**) is a small ∞ -category with finite products. A **model** (or **algebra**) for an ∞ -algebraic theory $\mathcal{A}$ is a functor $\mathcal{A} \rightarrow \mathcal{S}$ that preserves products.

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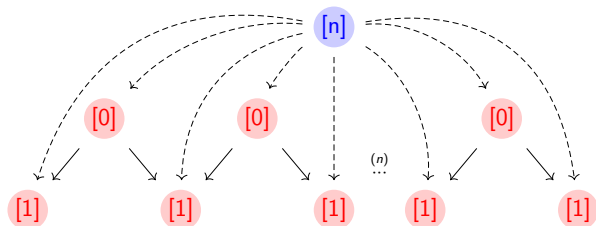
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Theorem (Rosicky 2007 and Lurie 2009)

The ∞ -category of models of an ∞ -algebraic theory is presentable.

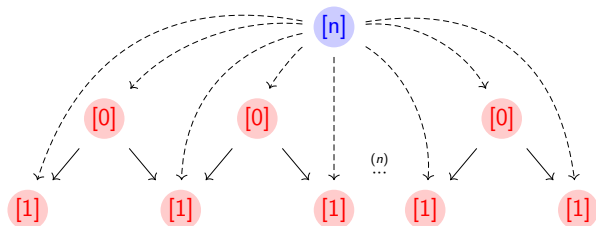
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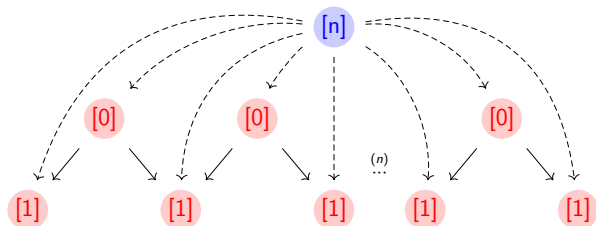


Then $\mathcal{T} = (\mathcal{A}, \{c_n \mid n \in \mathbb{N}\})$ is a limit ∞ -sketch, and a model $F : \mathcal{A} \rightarrow \mathcal{C}$ is a simplicial object in $\mathcal{C}$ such that

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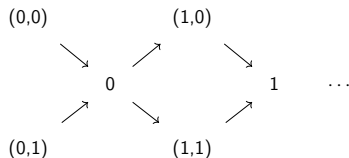
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Therefore, $\text{Mod}(\mathcal{T}, \mathcal{C})$ is the ∞ -category of **pre-category objects** in $\mathcal{C}$, and $\text{Mod}(\mathcal{T})$ is the ∞ -category of **Segal spaces**.

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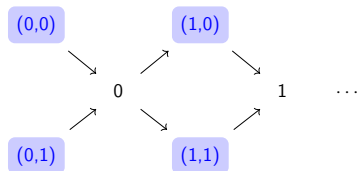
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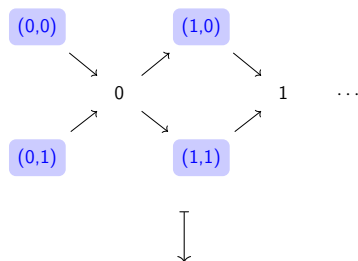
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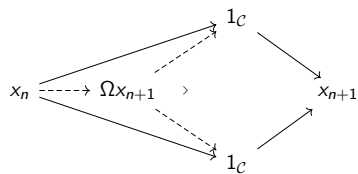
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A model for $\mathcal{T}$ is a sequence of pointed objects $1 \rightarrow x_n$ for every $n \in \mathbb{N}$, and maps $x_n \rightarrow \Omega x_{n+1}$.



$\text{Mod}(\mathcal{T}, \mathcal{C}) \simeq$ **Pre-spectrum objects**

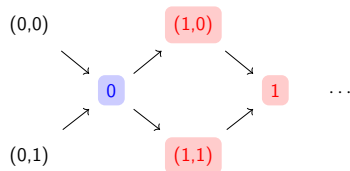
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Examples: Spectrum objects

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Let d_n be the cone with
diagram and **apex**. Then

$\mathcal{L}' = \mathcal{L} \sqcup \{d_n \mid n \in \mathbb{N}\}$, and
 $\mathcal{T}' = (\mathcal{A}, \mathcal{L}')$ is an ∞ -sketch.



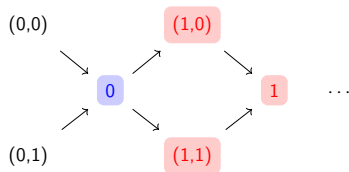
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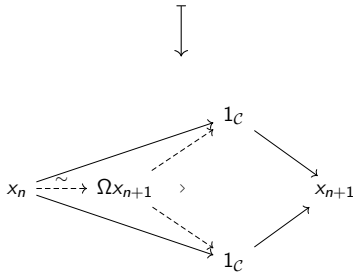


A model for $\mathcal{T}'$ is a pre-spectrum object in $\mathcal{C}$ such that

$$x_n \simeq \Omega x_{n+1}.$$

$$\text{Mod}(\mathcal{T}, \mathcal{C}) \simeq \mathbf{Spectrum\ objects}$$

$$\text{Mod}(\mathcal{T}) \simeq \mathbf{Spectra}$$



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$\implies$ By Lurie, we know that a presentable ∞ -category is equivalent to the ∞ -category of Ind-objects for some small ∞ -category. Then, use this characterization to construct a limit ∞ -sketch.

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Idea of the proof

- $\implies$ By Lurie, we know that a presentable ∞ -category is equivalent to the ∞ -category of Ind-objects for some small ∞ -category. Then, use this characterization to construct a limit ∞ -sketch.
- $\impliedby$ Prove that a model of any limit ∞ -sketch is equivalent to an M -local object for a set of morphisms M in the presentable ∞ -category $\mathrm{Fun}(\mathcal{A}^{\mathrm{op}}, \mathcal{S})$. Then, by Lurie, it is presentable.

Future work

- 💡 Proof the **model categorical version** of the representation theorem, replacing presentable ∞ -categories by combinatorial model categories.

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- 💡 **Generalization:** A ∞ -category is accessible if, and only if, it is equivalent to the ∞ -category of models of an ∞ -sketch.

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- 💡 Accessibility, presentability, sketches, and representation theorem for ∞ -**cosmoi** (Riehl-Verity 2022)

$\implies$ **Model-independent** version of this presentation!

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Thank you for listening!

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LIMIT-SKETCHABLE INFINITY CATEGORIES

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